

6G5Z0050 : Maths for Data Science

DURATION : 2 hours

Instructions to candidates:

Answer all the questions.

You must show all working to gain full marks.

Permitted materials:

Calculators are permitted. You may bring one double-sided A4 sheet of notes into the exam with you. A standard formula sheet will also be provided.

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1. (a) Students who took a test were asked to submit their marks for research purposes. 100 students responded to the call, and their marks are recorded and summarised in the grouped frequency table below.

Mark Range	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	8	18	25	22	16	6	3	2

- (i) Determine an estimate of the mean and standard deviation of the marks obtained. (7 marks)
- (ii) Explain why, given this data, you can only estimate the mean, and not calculate it precisely. (2 marks)

- (b) Considering the source of this data, how might you expect it to be biased? (2 marks)

- (c) The actual distribution of marks for this cohort of 500 students is given below. Perform a statistical test at the 5% significance level to determine how good the fit is for this data. State your hypotheses and conclusion.

Mark Range	10–19	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	40	95	125	110	90	26	10	4

(15 marks)

Total marks for question 1: [26]

2. (a) Manchester City and Manchester United play derby matches against each other roughly once a year, and of the 198 matches played so far, Manchester United has won 81. Assume the outcome of successive derbies are independent.
- (i) Let U represent the number of United wins over the next 40 derby matches. Use the historical data to write down a model U_B that represents U using a binomial distribution. State the requirements for a binomial distribution and explain why this is a suitable model. (7 marks)
 - (ii) Write down an approximation to U using a normal distribution. (4 marks)
 - (iii) Give one reason why this approximation is valid. (2 marks)
 - (iv) Calculate the probability that Manchester United will win more than half of the next 40 derby matches. (5 marks)
- (b) The mean attendance at football matches at Old Trafford is 73,979, and is normally distributed. A sample of 7 recent Manchester derby matches taking place at Old Trafford was collected, and the mean was found to be 74,444 and the standard deviation 955.4.
- (i) What type of statistical test should you use to test whether the mean attendance at derby games is different from the mean attendance in general? Justify your answer. (3 marks)
 - (ii) Perform this test at the 5% significance level, and explain the result, stating your hypotheses clearly. (10 marks)

Total marks for question 2: [31]

3. (a) (i) Find the determinant of the matrix $A = \begin{pmatrix} 6 & 2 \\ -5 & -2 \end{pmatrix}$ (4 marks)
- (ii) Calculate or write down the determinant of A^{-1} . (4 marks)
- (iii) Find the inverse of the matrix $B = \begin{pmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 1 & 4 & 4 \end{pmatrix}$ (10 marks)

- (b) Here is the covariance matrix for three variables, X , Y and Z :

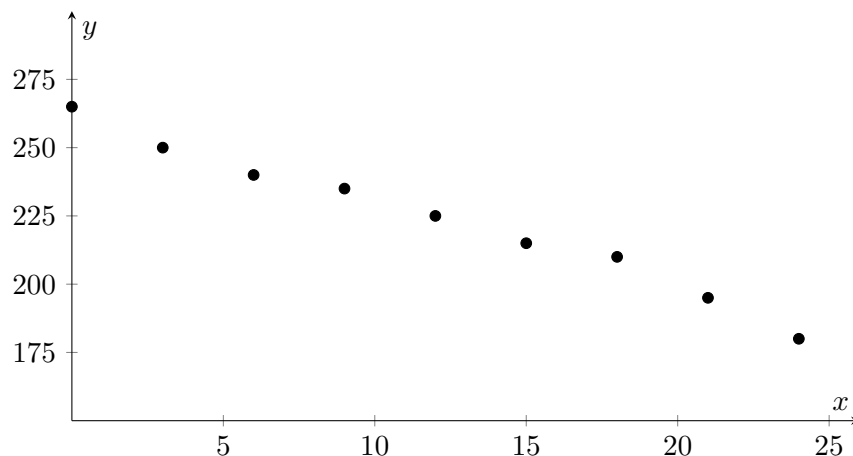
$$\begin{pmatrix} 60 & 32 & -4 \\ 32 & 30 & 0 \\ -4 & 0 & 80 \end{pmatrix}$$

Comment on the entries in this matrix - state what they represent, and what this tells you about the variables. (5 marks)

Total marks for question 3: [23]

4. (a) A new car tyre is fitted to a wheel. The tyre is inflated to its recommended pressure of 265 kPa and the wheel is left unused. At 3-month intervals, the tyre pressure is measured:

Time after fitting (x months)	0	3	6	9	12	15	18	21	24
Tyre Pressure (y kPa)	265	250	240	235	225	215	210	195	180



- (i) For this data,

$$\sum (x - \bar{x})(y - \bar{y}) = -1755$$

Calculate the equation of the least squares regression line of y on x.

(10 marks)

- (ii) Based on this equation, and confirming using the plot, what type of correlation is present between the variables?

(3 marks)

- (iii) Interpret in context the meaning of the value for the gradient of your regression line, and the approximate meaning of the value for the y-intercept.

(4 marks)

- (b) Is a linear model suitable for this situation? Explain why.

(3 marks)

Total marks for question 4: [20]

1 Reference - Statistical Tables

1.1 Chi-Squared Test Probability Table

DF	1.00	0.99	0.98	0.95	0.90	0.75	0.50	0.25	0.10	0.05	0.03	0.01	0.01
1	0.0000	0.0002	0.0010	0.0039	0.0158	0.1015	0.4549	1.3233	2.7055	3.8415	5.0239	6.6349	7.8794
2	0.0100	0.0201	0.0506	0.1026	0.2107	0.5754	1.3863	2.7726	4.6052	5.9915	7.3778	9.2103	10.597
3	0.0717	0.1148	0.2158	0.3519	0.5844	1.2125	2.3660	4.1083	6.2514	7.8147	9.3484	11.345	12.838
4	0.2070	0.2971	0.4844	0.7107	1.0636	1.9226	3.3567	5.3853	7.7794	9.4877	11.143	13.277	14.860
5	0.4117	0.5543	0.8312	1.1455	1.6103	2.6746	4.3515	6.6257	9.2364	11.071	12.833	15.086	16.750
6	0.6757	0.8721	1.2373	1.6354	2.2041	3.4546	5.3481	7.8408	10.645	12.592	14.449	16.812	18.548
7	0.9893	1.2390	1.6899	2.1674	2.8331	4.2549	6.3458	9.0372	12.017	14.067	16.013	18.475	20.278
8	1.3444	1.6465	2.1797	2.7326	3.4895	5.0706	7.3441	10.219	13.362	15.507	17.535	20.090	21.955
9	1.7349	2.0879	2.7004	3.3251	4.1682	5.8988	8.3428	11.389	14.684	16.919	19.023	21.666	23.589
10	2.1559	2.5582	3.2470	3.9403	4.8652	6.7372	9.3418	12.549	15.987	18.307	20.483	23.209	25.188
11	2.6032	3.0535	3.8158	4.5748	5.5778	7.5841	10.341	13.701	17.275	19.675	21.920	24.725	26.757
12	3.0738	3.5706	4.4038	5.2260	6.3038	8.4384	11.340	14.845	18.549	21.026	23.337	26.217	28.300
13	3.5650	4.1069	5.0088	5.8919	7.0415	9.2991	12.340	15.984	19.812	22.362	24.736	27.688	29.819
14	4.0747	4.6604	5.6287	6.5706	7.7895	10.165	13.339	17.117	21.064	23.685	26.119	29.141	31.319
15	4.6009	5.2294	6.2621	7.2609	8.5468	11.037	14.339	18.245	22.307	24.996	27.488	30.578	32.801
16	5.1422	5.8122	6.9077	7.9617	9.3122	11.912	15.339	19.369	23.542	26.296	28.845	32.000	34.267
17	5.6972	6.4078	7.5642	8.6718	10.085	12.792	16.338	20.489	24.769	27.587	30.191	33.409	35.718
18	6.2648	7.0149	8.2308	9.3905	10.865	13.675	17.338	21.605	25.989	28.869	31.526	34.805	37.156
19	6.8440	7.6327	8.9065	10.117	11.651	14.562	18.338	22.718	27.204	30.144	32.852	36.191	38.582
20	7.4338	8.2604	9.5908	10.851	12.443	15.452	19.337	23.828	28.412	31.410	34.170	37.566	39.997
21	8.0337	8.8972	10.283	11.591	13.240	16.344	20.337	24.935	29.615	32.671	35.479	38.932	41.401
22	8.6427	9.5425	10.982	12.338	14.041	17.240	21.337	26.039	30.813	33.924	36.781	40.289	42.796
23	9.2604	10.196	11.689	13.091	14.848	18.137	22.337	27.141	32.007	35.172	38.076	41.638	44.181
24	9.8862	10.856	12.401	13.848	15.659	19.037	23.337	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	19.939	24.337	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	20.843	25.336	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	21.749	26.336	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	22.657	27.336	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	23.567	28.336	33.711	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	24.478	29.336	34.800	40.256	43.773	46.979	50.892	53.672

1.2 T-Test Probability Table

1-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.03	0.01	0.005	0.001	0.0005
2-tail	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.010	0.002	0.001
1	0.000	1.000	1.376	1.963	3.078	6.314	12.710	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

1.3 Normal Distribution - Probability Table

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							